

DEVOPS MASTER COURSE

Hands-On Labs, Assessments and Capstone

Executable practice system, scoring model and enterprise graduation project

Edition	2026 Final Release
Curriculum	Volumes 1-15
Delivery	DOCX + PDF + Executable Repository

Professional learning, practical evidence, enterprise readiness.

Executive Summary

This manual operationalizes the curriculum. It defines the 150-lab repository, assessment architecture, evidence requirements, interview simulation model, and the final enterprise capstone. Every practical exercise follows a build-test-fail-recover-document-cleanup loop.

1. Lab Repository Design

Component	Implemented quantity	Purpose
Volume lab folders	15	One practical track per curriculum volume
Structured lab guides	150	Ten production-style labs per volume
Assessment banks	15	Knowledge, scenario and practical examinations
Starter kits	4	Containers, Kubernetes, Terraform and CI
Reusable templates	3+	Lab evidence, postmortem and runbook
Automated scripts	3	Structure validation, listing and file audit

2. Standard Lab Lifecycle

1. Define the business scenario and measurable outcome.
2. Record the starting state, versions and constraints.
3. Implement the smallest safe solution.
4. Add repeatable validation and security checks.
5. Inject a realistic failure.
6. Diagnose from observable evidence rather than assumptions.
7. Recover through rollback, restore or remediation.
8. Document trade-offs and enterprise-scale changes.
9. Remove temporary resources and prove cleanup.

3. Lab Scoring Rubric

Dimension	Weight	Pass evidence
Functional correctness	25%	Expected behavior and automated validation
Reliability and recovery	20%	Failure drill, diagnosis and successful recovery
Security	15%	Least privilege, secret protection and vulnerability checks
Automation and reproducibility	15%	Versioned code and repeatable execution
Observability and evidence	15%	Logs, metrics, traces, outputs and diagrams
Documentation and cleanup	10%	Runbook, reflection and no residual resources

Passing threshold: 70% overall, with no zero score in security, reliability, or cleanup.

4. Assessment System

Assessment	Frequency	Format	Threshold
Chapter knowledge check	Every chapter	5-10 questions	70%
Part practical exam	Every part	Timed implementation	70%
Volume scenario review	Every volume	Architecture and incident scenario	75%
Mock interview	Volumes 6-15	Technical and behavioral	Professional rating
Final capstone defense	Course completion	Demo, design review and incident drill	80%

5. Practical Examination Template

- Receive a production scenario, constraints and target outcome.
- Create a brief plan and risk assessment before changing the environment.
- Implement the solution without copying a complete answer.
- Demonstrate tests, logs, failure handling and rollback.
- Explain two rejected alternatives and why they were rejected.
- Submit evidence and a concise operational handover.

6. Interview Evaluation Model

Competency	Weak	Professional	Master
Problem framing	Jumps to tools	Clarifies requirements	Frames business, risk and constraints
Technical depth	Recites commands	Explains architecture and failures	Connects layers and trade-offs
Troubleshooting	Random trial and error	Uses hypotheses and evidence	Stabilizes, delegates and prevents recurrence
Communication	Unstructured	Clear sequence and rationale	Adapts detail to engineers and executives
Ownership	Stops after fix	Documents and verifies	Drives systemic improvement

7. Final Enterprise Capstone

The capstone integrates the complete curriculum into a single enterprise delivery platform. The learner builds a sample application and the platform needed to develop, secure, deploy, observe, operate and improve it.

Workstream	Required implementation
Source and collaboration	Protected branches, pull requests, reviews and release tags
CI/CD	Test, build, scan, sign, publish, deploy and rollback
Infrastructure	Versioned IaC, environment separation and state protection
Runtime	Containers and Kubernetes with health, scaling and policy
Security	Secrets, least privilege, SBOM, provenance and policy gates

Workstream	Required implementation
Observability	Metrics, logs, traces, dashboards and actionable alerts
SRE	SLIs, SLOs, error budget, runbooks and incident command
Platform engineering	Golden path and onboarding for a second service
Cloud and FinOps	Budget, tagging, capacity and teardown evidence
Leadership	Architecture decision record, risk register and executive summary

8. Mandatory Failure Drills

Scenario	Expected learner response
Failed deployment	Detect, stabilize, roll back and identify failed gate
Dependency latency	Correlate traces, metrics and logs; apply graceful degradation
Credential exposure	Revoke, rotate, investigate scope and prevent recurrence
Node or zone failure	Verify rescheduling, availability and capacity
Telemetry outage	Operate safely with degraded observability and restore pipeline
Data restore	Recover to target RPO and verify integrity
Cost anomaly	Identify driver, contain spend and preserve service objectives

9. Capstone Delivery Phases

Phase	Output	Quality gate
1. Discovery	Requirements, architecture and risks	Review approved
2. Foundation	Repository, IaC and environments	Reproducible bootstrap
3. Delivery	CI/CD and runtime platform	Safe release and rollback
4. Assurance	Security and supply-chain controls	Policy gates pass
5. Reliability	Observability, SLOs and runbooks	Failure drills pass
6. Platform product	Golden path and second service	Onboarding demonstrated
7. Graduation	Portfolio, demo and defense	80%+ final score

10. Certification and Completion Rules

- Completion certificates should represent demonstrated competence, not attendance alone.
- The learner must retain evidence for every required practical exam.
- Any critical security or cleanup failure requires reattempt.
- Capstone acceptance requires both technical demonstration and oral defense.
- The final portfolio must include architecture, code, evidence, incident learning and outcomes.

Appendix A - Repository Commands

```
cd devops-master-course-labs
make validate
make list
python scripts/audit_course_files.py --root /path/to/course/files
```

Appendix B - Evidence Naming Standard

```
evidence/
├─ 01-environment-inventory.txt
├─ 02-architecture.png
├─ 03-commands-and-config.md
├─ 04-validation-results.txt
├─ 05-failure-drill-timeline.md
├─ 06-recovery-proof.txt
└─ 07-cleanup-proof.txt
```

Appendix C - Volume Lab Map

Volume	Specialization	Repository focus
1	Linux and DevOps Foundations	Linux administration and secure operations
2	Shell and Troubleshooting	Automation, diagnostics and recovery
3	Git and Collaboration	Branching, reviews and release workflow
4	Networking and Web	DNS, TLS, routing and web services
5	Python Automation	CLI, API and testable automation
6	Docker	Images, Compose, security and operations
7	Kubernetes	Workloads, networking, storage and RBAC
8	IaC and Configuration	Terraform, Ansible, policy and drift
9	CI/CD Foundations	Pipelines, artifacts and release controls
10	GitOps and DevSecOps	Progressive delivery and supply-chain assurance
11	Observability and SRE	Telemetry, SLOs and incident response
12	Enterprise Cloud	AWS, Azure, GCP, hybrid and FinOps
13	Platform Engineering	Portals, self-service and golden paths
14	MLOps and AI Infrastructure	Data, models, serving, RAG and AI operations
15	Leadership and Mastery	Transformation, governance and executive simulations